

# Energy stores and systems

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

|    | QUESTION                                                                            | ANSWER                                                                                                      | ✓                        |
|----|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------|
| 1  | Name the eight stores of energy.                                                    | Chemical, electrostatic, magnetic, elastic potential, nuclear, thermal, kinetic and gravitational potential | <input type="checkbox"/> |
| 2  | State the law of conservation of energy.                                            | Energy can be transferred usefully, stored or dissipated, but it cannot be created or destroyed.            | <input type="checkbox"/> |
| 3  | Define a closed system.                                                             | A system that no energy enters or leaves.                                                                   | <input type="checkbox"/> |
| 4  | Define work done.                                                                   | The energy transferred when a force moves an object through a distance.                                     | <input type="checkbox"/> |
| 5  | State the SI unit of energy.                                                        | Joule (J)                                                                                                   | <input type="checkbox"/> |
| 6  | Name the store of energy associated with fuels and batteries.                       | The chemical energy store                                                                                   | <input type="checkbox"/> |
| 7  | Describe the change in the chemical energy store as fuel is burned.                 | It decreases.                                                                                               | <input type="checkbox"/> |
| 8  | Name the store of energy associated with changes in speed.                          | The kinetic energy store                                                                                    | <input type="checkbox"/> |
| 9  | Describe the change in the kinetic energy store as an object slows down.            | It decreases.                                                                                               | <input type="checkbox"/> |
| 10 | Name the store of energy associated with changes in height.                         | The gravitational potential energy store                                                                    | <input type="checkbox"/> |
| 11 | Describe the change in the gravitational potential energy store as an object rises. | It increases.                                                                                               | <input type="checkbox"/> |
| 12 | Name the store of energy associated with extending and compressing objects.         | The elastic potential energy store                                                                          | <input type="checkbox"/> |

| QUESTION |                                                                                                           | ANSWER                                                                                                                    | ✓                        |
|----------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 13       | Describe the change in the elastic potential energy store as an object is compressed more.                | It increases.                                                                                                             | <input type="checkbox"/> |
| 14       | Name the store of energy associated with changes in temperature.                                          | The thermal energy store                                                                                                  | <input type="checkbox"/> |
| 15       | Describe the change in the thermal energy store as an object gets hotter.                                 | It increases.                                                                                                             | <input type="checkbox"/> |
| 16       | Name the four ways energy is transferred between stores.                                                  | Mechanically, electrically, by heating and by radiation                                                                   | <input type="checkbox"/> |
| 17       | Name the energy transfer when a force moves an object.                                                    | Mechanical transfer                                                                                                       | <input type="checkbox"/> |
| 18       | Name the energy transfer when an object emits light or sound.                                             | Transfer by radiation                                                                                                     | <input type="checkbox"/> |
| 19       | Name the energy transfer when a hot object warms a cooler one.                                            | Transfer by heating                                                                                                       | <input type="checkbox"/> |
| 20       | Define a system.                                                                                          | An object or a group of objects.                                                                                          | <input type="checkbox"/> |
| 21       | State what happens to energy when a system changes.                                                       | It is transferred from one store to another.                                                                              | <input type="checkbox"/> |
| 22       | State the equation linking kinetic energy, mass and speed.                                                | kinetic energy = $0.5 \times \text{mass} \times \text{speed}^2$<br>( $E_k = \frac{1}{2} m v^2$ )                          | <input type="checkbox"/> |
| 23       | State the equation linking gravitational potential energy, mass, gravitational field strength and height. | gravitational potential energy = mass $\times$ gravitational field strength $\times$ height<br>( $E_p = m g h$ )          | <input type="checkbox"/> |
| 24       | State the equation linking elastic potential energy, spring constant and extension.                       | elastic potential energy = $0.5 \times \text{spring constant} \times \text{extension}^2$<br>( $E_e = \frac{1}{2} k e^2$ ) | <input type="checkbox"/> |
| 25       | Name the two quantities that kinetic energy depends on.                                                   | Mass and speed                                                                                                            | <input type="checkbox"/> |
| 26       | Name the three quantities that gravitational potential energy depends on.                                 | Mass, gravitational field strength and height                                                                             | <input type="checkbox"/> |

| QUESTION |                                                                                             | ANSWER                                                                             | ✓                        |
|----------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------|
| 27       | <b>A ball is thrown straight up. Describe the changes in its energy stores as it rises.</b> | Kinetic energy store decreases;<br>gravitational potential energy store increases. | <input type="checkbox"/> |
| 28       | <b>An electric kettle is switched on. Describe the useful energy transfer.</b>              | Energy is transferred electrically to the thermal energy store of the water.       | <input type="checkbox"/> |